

MASTER PROJECT INVENTORY & TECHNICAL SPECIFICATIONS

Candidate & Author: Thabeeb Jafran — Business Analyst | Data Analyst | Project Manager
Target Domain: Data Analytics, Business Intelligence, Data Engineering & Agile Project Management
Document Purpose: Exhaustive technical and strategic reference inventory compiled directly from deep code-level repositories, raw data schemas, statistical models, and workflow specifications.

SECTION 1: EXECUTIVE PORTFOLIO SUMMARY & CORE COMPETENCIES

Thabeeb Jafran brings over 5 years of experience across Fintech, E-commerce, Logistics, and Retail Analytics. The portfolio demonstrates an end-to-end capability lifecycle: from initial data ingestion, schema design, and cleaning to predictive machine learning, statistical hypothesis testing, interactive BI reporting, and financial ROI risk modeling.

Core Technical & Strategic Stack

Data Analysis & Engineering: SQL (PostgreSQL, MySQL), Python (

```
pandas
,
numpy
,
scikit-learn
,
statsmodels
,
geopy
,
seaborn
,
matplotlib
```

-), Data Modeling (Star/Snowflake Schemas, Data Vault basics).
- **Business Intelligence & Visualization:** Power BI (DAX, Dataflows, Power Query), Tableau (Tableau Desktop, Calculation Fields, LOD Expressions, Tableau Public), Gamma.
- **Statistical Analysis & Econometrics:** One-Way ANOVA, Post-hoc Pairwise T-Tests (with Bonferroni Correction), Linear Regression, Logistic Regression, Feature Importance, Outlier Capping (IQR Method), RFM Segmentation, Cohort Retention Analysis.
- **Workflow Automation & AI Orchestration:** n8n, AI Agent Chains (Google Gemini API), JSON Output Enforcement, Power Automate.
- **Project & Operations Management:** Agile/Scrum, Waterfall/Predictive Hybrid, Jira, Asana, Work Breakdown Structure (WBS), ROAM Risk Management, PERT Three-Point Schedule Estimation, EAC/ETC Cost Forecasting, EMV Risk Analysis.

SECTION 2: DEEP TECHNICAL SPECIFICATIONS BY REPOSITORY

1. Credit Card Financial Dashboard (

credit-card-financial-dashboard

)

- **Domain:** Fintech / Banking Analytics
- **Business Context:** A credit card issuer needed real-time tracking of weekly revenue, transaction channels, customer segments, delinquency rates, and customer acquisition costs across card tiers to eliminate manual, delayed reporting.

Key Metrics & Impact:

- **Dataset Volume:** 656,000+ transactions across 10,294 unique customers (\$55.3M total revenue).
- **Revenue Drivers:** Blue & Silver card tiers drive **93%+** of total portfolio revenue.
- **Demographic & Channel Breakdowns:** Male customers generated **\$30M** vs. **\$25M** (Female); **Swipe** generated **\$35M**, **Chip** **\$17M**, **Online** **\$3M**.
- **Geographic Focus:** TX, NY, and CA combined account for **~68%** of total revenue.
- **Portfolio Health:** **57.5%** activation rate, **6.06%** delinquency rate (highest in self-employed & business owner segments).

- **Trend Detection:** Week 52 flagged an immediate **12.8% Week-over-Week revenue decline** (reversing a +4.3% prior week gain).

Technical Architecture & Data Pipeline

Data Ingestion: Designed PostgreSQL schema with dual transactional and incremental load tables (

cc_detail
,
cust_detail
,
cc_add
,
cust_add

1.) to simulate continuous data pipeline refreshes.

2. **Database Schema (PostgreSQL):**

```
CREATE DATABASE ccdb;

CREATE TABLE cc_detail (
  Client_Num INT,
  Card_Category VARCHAR(20),
  Annual_Fees INT,
  Activation_30_Days INT,
  Customer_Acq_Cost INT,
  Week_Start_Date DATE,
  Week_Num VARCHAR(20),
  Qtr VARCHAR(10),
  current_year INT,
  Credit_Limit DECIMAL(10,2),
  Total_Revolving_Bal INT,
  Total_Trans_Amt INT,
  Total_Trans_Ct INT,
  Avg_Utilization_Ratio DECIMAL(10,3),
  Use_Chip VARCHAR(10),
  Exp_Type VARCHAR(50),
  Interest_Earned DECIMAL(10,3),
  Delinquent_Acc VARCHAR(5)
);
```

```

CREATE TABLE cust_detail (
    Client_Num INT,
    Customer_Age INT,
    Gender VARCHAR(5),
    Dependent_Count INT,
    Education_Level VARCHAR(50),
    Marital_Status VARCHAR(20),
    State_cd VARCHAR(50),
    Zipcode VARCHAR(20),
    Car_Owner VARCHAR(5),
    House_Owner VARCHAR(5),
    Personal_Loan VARCHAR(5),
    Contact VARCHAR(50),
    Customer_Job VARCHAR(50),
    Income INT,
    Cust_Satisfaction_Score INT
);

SET datestyle = 'DMY';

COPY cc_detail
FROM 'E:\Business Analytics\Power BI\Projects\credit_card.csv'
DELIMITER ',' CSV HEADER;

COPY cust_detail
FROM 'E:\Business Analytics\Power BI\Projects\customer.csv'
DELIMITER ',' CSV HEADER;

```

1. Power BI DAX Calculated Columns & Measures:

```

-- Age Group Classification
AgeGroup =
SWITCH(
    TRUE(),
    'public cust_detail'[customer_age] < 30, "20-30",
    'public cust_detail'[customer_age] >= 30 && 'public cust_detail'[customer_age] < 40, "30-40",
    'public cust_detail'[customer_age] >= 40 && 'public cust_detail'[customer_age] < 50, "40-50",
    'public cust_detail'[customer_age] >= 50 && 'public cust_detail'[customer_age] < 60, "50-60",
    'public cust_detail'[customer_age] >= 60, "60+",
    "unknown"
)

-- Income Group Classification
IncomeGroup =
SWITCH(
    TRUE(),
    'public cust_detail'[income] < 35000, "Low",
    'public cust_detail'[income] >= 35000 && 'public cust_detail'[income] < 70000, "Med",
    'public cust_detail'[income] >= 70000, "High",
    "unknown"
)

-- Week Number Generation
week_num2 = WEEKNUM('public cc_detail'[week_start_date])

-- Total Revenue Formula
Revenue =

```

```

'public cc_detail'[annual_fees] +
'public cc_detail'[total_trans_amt] +
'public cc_detail'[interest_earned]

-- Current Week Revenue Measure
Current_week_Revenue =
CALCULATE(
    SUM('public cc_detail'[Revenue]),
    FILTER(
        ALL('public cc_detail'),
        'public cc_detail'[week_num2] = MAX('public cc_detail'[week_num2])
    )
)

-- Previous Week Revenue Measure
Previous_week_Revenue =
CALCULATE(
    SUM('public cc_detail'[Revenue]),
    FILTER(
        ALL('public cc_detail'),
        'public cc_detail'[week_num2] = MAX('public cc_detail'[week_num2]) - 1
    )
)

```

2. TalentGuard — IBM Employee Attrition Prediction & Retention Strategy (

talentguard-ibm-employee-attrition-prediction

)

- **Domain:** HR Analytics / Predictive Machine Learning
- **Business Context:** IBM experienced a 16.14% attrition rate in 2022. HR needed a transition from reactive exit interviews to proactive ML risk scoring to allocate retention budgets effectively.

Key Metrics & Impact:

- **Dataset:** 1,470 employee records (35 features including compensation, role, tenure, overtime, satisfaction).
- **Model Performance:** Evaluated via Confusion Matrix, Classification Report, ROC Curve, and **KS-Statistic = 38.9** (indicating strong separation power between leavers and stayers).
- **Baseline Flagging:** 1,205 active employees evaluated; decision threshold tuned down to **0.3** (surfacing 148 active employees / 12.2% at high risk).
- **Primary Drivers:** **100%** of predicted leavers worked Overtime; **67.6%** earned <\$5,000/month; **52.7%** held zero stock options.
- **Simulation Impact:** Strategy simulation dropped predicted attrition from **12.2% down to 8.8%** (-3.4 percentage points overall), reduced Overtime-linked churn by **-28.3%**, and reduced No-Stock-Option churn by **-55.1%**.

Machine Learning Implementation (

Python

)

Feature Selection: Filtered using Variance Inflation Factor (VIF) to eliminate multicollinearity, selecting top actionable features:

MonthlyIncome

,

OverTime

, and

StockOptionLevel

- .

- **Model Script Framework:**

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Data Load & Preprocessing
ibm = pd.read_csv('ibm.csv')
ibm['Attrition_Num'] = ibm['Attrition'].apply(lambda x: 1 if x == 'Yes' else 0)
ibm['OverTime_Num'] = ibm['OverTime'].apply(lambda x: 1 if x == 'Yes' else 0)

# VIF Analysis & Logistic Regression
X = ibm[['MonthlyIncome', 'OverTime_Num', 'StockOptionLevel']]
y = ibm['Attrition_Num']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression()
model.fit(X_train, y_train)

# Probability Scoring with custom threshold = 0.3
probs = model.predict_proba(X_test)[: , 1]
preds = (probs >= 0.3).astype(int)
```

3. SmartFare NYC — Uber Pricing Optimization & Revenue Analysis (

smartfare-nyc-uber-pricing-optimization

)

- **Domain:** Transportation / Econometric Modeling & Dynamic Pricing
 - **Business Context:** Uber NYC experienced a **9.5% revenue decline** between Q1 2013 and Q2 2015, alongside a **14.2% drop in trip volume**. Objective: Design an optimized, dynamic surge/discount pricing model targeting a **$\geq 10\%$ revenue lift** without increasing base fares.
- Key Metrics & Impact:**
- **Data Volume:** Cleaned 64,127 NYC trip records from 200,000 raw rows (removing invalid coordinates, out-of-bounds bounding boxes, zero/negative fares).
 - **Regression Performance:** Multiple Linear Regression model explained **71.8% of fare variance** ($R^2 = 0.718$).
 - **Key Variable Weight:** Trip Distance was the dominant driver (standardized coefficient $\beta = 0.86$).
 - **Financial Result:** Back-tested dynamic tiered pricing model produced a **+11.5% revenue increase** (\$617K to \$689K in modeled sample) while keeping base fares intact.

Statistical & Technical Workflow

Feature Engineering: Extracted geodesic trip distances via

geopy

1. , 1km grid pickup zones aggregated to boroughs, 3-hour time-of-day buckets, AM/PM splits, and passenger count categorizations.

IQR Outlier Trimming: Filtered extreme values on

fare_amount

and

distance

2. .

3. **Optimized Tiered Pricing Logic:** Applied \$\pm 20\%\$ surge/discount adjustments based on peak hour density, borough demand ratios, and group sizes (5-6 passengers).

4. Retail UK — E-Commerce Cohort Analysis & RFM Segmentation (

retail-uk-cohort-analysis-rfm-segmentation

)

- **Domain:** E-Commerce / Customer Retention Analytics
- **Business Context:** By Nov 2011, 8 out of 11 monthly customer cohorts fell below the company's 31% retention benchmark. Needed an exact RFM behavioral segmentation and Customer Retention Cost (CRC) spend threshold.

Key Metrics & Impact:

- **Segmentation:** Identified 7 high-value actionable customer segments (3,089 total customers). Identified the largest single segment as "Loss" (RFM 111 — lapsed, low frequency, low spend) and excluded it from retention budgets.
- **Financial CRC Model:** Calculated target Customer Retention Cost (CRC) range of **\$3.80 – \$15.20 per customer** based on CLTV cohort economics to guarantee CLTV-positive marketing campaigns.

Data Cleaning & Execution Pipeline (

Python

)

- Dropped records missing

CustomerID

, removed duplicates, filtered out negative/zero quantities and unit prices (returns/errors), and applied IQR-based outlier capping.

5. FastFood US — Marketing Campaign A/B Testing (

fastfood-us-ab-testing-campaign-analysis

)

- **Domain:** Retail Marketing / Hypothesis Testing
- **Business Context:** Fast food franchise network tested 3 distinct marketing campaigns across 137 sample stores (spread across Large, Medium, Small markets) to determine which campaign statistically drives higher revenue.

Key Metrics & Impact:

- **Sample Distribution:** Campaign 1 (43 stores / 31.4%), Campaign 2 (47 stores / 34.3%), Campaign 3 (47 stores / 34.3%).

ANOVA Results: Significant campaign impact across all market tiers:

- Full Sample ANOVA: $p = 0.0037$
- Large Market ANOVA: $p = 0.0077$
- Medium Market ANOVA: $p = 0.0001$
- Small Market ANOVA: $p = 0.0032$
- **Post-hoc T-Tests & Bonferroni Correction:** Applied adjusted significance threshold $\alpha_{adj} = 0.05 / 3 = 0.0167$.
- **Conclusion:** Campaign 1 and Campaign 3 are statistically tied and significantly outperform Campaign 2 across all markets. Recommended retiring Campaign 2 entirely.

6. Customer Shopping Behavior Analysis (

customer-shopping-behavior-analysis

)

- **Domain:** Consumer Retail & Behavior Analytics
- **Business Context:** Analyzing 3,900 retail transactions to understand demographic spending patterns, discount dependencies, and subscription status influence.

Key Metrics & Impact:

- Male customers generated ~2x the total revenue of female customers.
- Non-subscribers contribute higher total revenue due to total volume despite similar average spend per transaction.
- 3,116 of 3,900 customers classified into "Loyal" segment (>10 previous purchases).

Full SQL Business Queries (

`customer-behavior-sql-queries.sql`

)

```
-- Q1. Total revenue generated by male vs. female customers
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;

-- Q2. Customers using discounts who spent more than average purchase amount
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);

-- Q3. Top 5 products with highest average review rating
SELECT item_purchased, ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;

-- Q4. Compare average Purchase Amounts between Standard and Express Shipping
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;

-- Q5. Spending comparison between Subscribers and Non-subscribers
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;

-- Q6. Top 5 products with highest percentage of discounted purchases
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;

-- Q7. Customer Segmentation (New, Returning, Loyal)
```

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
  CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
  END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;

-- Q8. Top 3 most purchased products per category using Window Functions
WITH item_counts AS (
  SELECT category,
    item_purchased,
    COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

-- Q9. Subscription distribution among repeat buyers (>5 previous purchases)
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;

-- Q10. Revenue contribution by age group
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

7. AI Job Application Assistant (

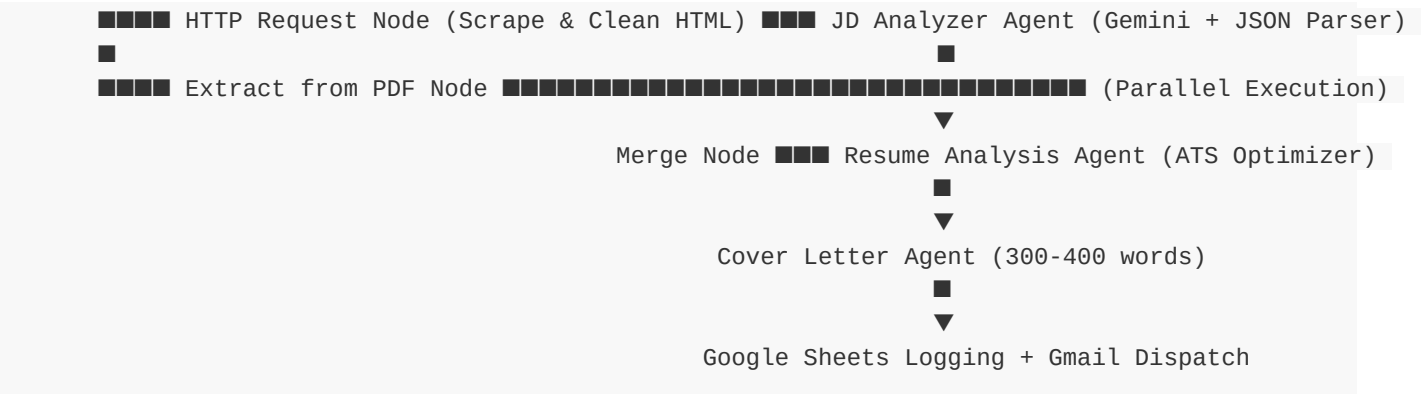
ai-job-application-assistant

)

- **Domain:** Workflow Automation / Generative AI Engineering
- **Business Context:** Built a zero-touch, automated multi-agent AI workflow in n8n powered by Google Gemini to analyze job descriptions, parse resumes, generate ATS optimization reports, write cover letters, and log data.
- **System Architecture & Agent Chain:**

Form Submission (Name, Email, PDF Resume, Job Posting URL)





- **Technical Implementation:** Enforced strict JSON schemas at the prompt level for each agent, allowing continuous data flow across n8n nodes without manual reformatting.

8. Mobile Banking Application — Financial Analysis (

mobile-banking-financial-analysis

)

- **Domain:** Project Financial Management & Regulatory Compliance Analysis
- **Business Context:** A 6-month mid-project regulatory compliance mandate required adding advanced security features to a \$500K mobile banking app project.

Financial Metrics & Formulas:

- **Budget Impact:** Cost expanded from \$500K to **\$600K (+20%)**; Timeline extended from **12 to 14 months**.
- **ROI Shift:** Baseline projected ROI of **+25%** shifted to **-20%**; Cost-Benefit Ratio (CBR) fell from 1.25 to **0.83**.

Expected Monetary Value (EMV) Risk Calculation: $\text{EMV} = \text{Probability of Regulatory Fine} \times \text{Financial Impact}$

- Without Security Feature: $\text{EMV} = 12\% \times (-\$600,000) = -\$72,000$.
 - With Security Feature: $\text{EMV} = 0.05\% \times (-\$600,000) = -\$300$.
 - **Outcome:** 240x risk exposure reduction, fully justifying the \$100K budget variance.
- **PERT Schedule Validation:** Validated revised 8-month remaining development phase using 3-point estimate: $T_{\text{expected}} = \frac{O + 4M + P}{6}$

9. AHI Marketing Analytics App (

ahi-marketing-analytics-app

)

- **Domain:** Agile / Predictive Project Management
- **Business Context:** Acme Health Innovations needed to replace fragmented manual systems with a real-time analytics app for internal teams and customers (\$250K fixed budget, 90-day timeline, 20% Year 1 ROI target).
- **Key Results:** Delivered within 90 days and within budget (\$27K approved additional equipment/data scope changes). Extended sprints from 2 to 3 to accommodate mid-sprint scope changes. Identified key infrastructure lesson: over-provisioned infrastructure for 20M users vs 4M actual target.

10. Plant Pals Operations Project (

plant-pals-operations-project

)

- **Domain:** Operations Management / B2B Logistics
- **Business Context:** Office Green launched a B2B plant subscription service targeting +5% company revenue growth and +10% customer retention.

Operational Achievements:

- Improved on-time deliveries from **80% to 90%** during pilot launch.
 - Increased customer satisfaction (CSAT) by **+42%**.
 - Reduced delivery unit costs by **-\$0.50/unit**.
 - Standardized potting procedures to eliminate a **10% mis-potting defect rate**.
 - Utilized **ROAM Risk Framework** (Resolved, Owned, Accepted, Mitigated) for risk tracking.
-

11. Sauce & Spoon Tablet Rollout (

sauce-spoon-tablet-rollout

)

- **Domain:** Retail Operations / Product Implementation
- **Business Context:** Tabletop menu tablet pilot across two restaurant locations (Downtown and North) to increase table turnover and guest throughput.
- **Key Results:** Sales grew **>20%**; Daily guest count increased **10%**; Wait times reduced by **30 minutes**; Checkout time cut to **<1 minute**; Food waste dropped **50%** (via order accuracy); CSAT increased from **72% to 86%**.

12. Virtual Verde Project (

virtual-verde-scrum-project

)

- **Domain:** Agile / Scrum Facilitation
- **Business Context:** Scrum delivery of e-commerce features (tutorials, help pages, vendor management system).
- **Key Results:** Completed 6 of 8 backlogged sprint items; managed mid-release vendor management system bug resolution without slipping primary release schedule.

13. Tableau Projects Portfolio (

tableau-projects-portfolio

)

- **Domain:** Visual Analytics / Portfolio

Overview: 20 interactive dashboards published on Tableau Public. Key technical highlights include:

1. **Video Game Sales:** Analyzed 11,493 titles across 31 platforms and 579 publishers.
2. **Superstore Sales:** Analyzed \$14.9M in transactional revenue across retail categories.

- 3. **UK Road Accidents:** Risk evaluation across 144,000+ accident records.
- 4. **Revenue Analysis Dashboard:** Enterprise breakdown of \$233M+ revenue by age, geography, and category.
- 5. **E-commerce YTD Sales:** \$14.1M YTD tracking with monthly cohort acquisition views.

SECTION 3: MASTER CONSOLIDATED METRICS REFERENCE TABLE

Project Name	Domain	Tools Used	Primary Metric / Outcome	Secondary Metric / Outcome
Credit Card Financial Dashboard	Fintech / BI	PostgreSQL, Power BI, DAX	\$55.3M Total Revenue tracked across 656K transactions	Blue & Silver tiers drive 93%+ revenue; 6.06% delinquency rate
TalentGuard IBM Attrition	HR / ML	Python, scikit-learn, Tableau	Reduced predicted attrition rate from 12.2% to 8.8% (-3.4 pts)	KS-Statistic of 38.9; 100% of predicted leavers worked Overtime
SmartFare NYC Uber Pricing	Econometrics	Python, statsmodels, Tableau	Projected +11.5% revenue lift (\$617K to \$689K sample)	Model R ² = 0.718; Distance std beta = 0.86
Retail UK RFM Cohort	E-commerce	Python, pandas, Cohorts	Target Customer Retention Cost (CRC) set to \$3.80 - \$15.20	Segmented 3,089 target customers; isolated "Loss" segment
FastFood US A/B Testing	Marketing	Excel, ANOVA, T-Tests	Proved Campaign 1 & 3 statistically tie and beat Campaign 2	Full Sample ANOVA p = 0.0037; Bonferroni threshold = 0.0167

Customer Behavior Analysis	Retail	SQL, Python, Power BI	Analyzed 3,900 rows; Male spend 2x Female spend	Classified 3,116 customers as "Loyal" segment
AI Job Application Assistant	AI / Automation	n8n, Google Gemini, JSON	100% end-to-end zero-touch workflow automation	3-agent chain enforcing strict structured JSON output
Mobile Banking Analysis	Finance / PM	EAC, EMV, PERT, Excel	Regulatory compliance EMV risk reduced 240x (-\$72K to -\$300)	Handled +20% budget shift (\$500K to \$600K)
AHI Marketing Analytics App	Agile PM	Jira, Scrum, Waterfall	Delivered within 90 days & \$250K budget	Shifted setup from 2 to 3 sprints; targeted 20% Year 1 ROI
Plant Pals Operations	Operations	Asana, Excel, ROAM	On-time delivery boosted from 80% to 90%	CSAT increased +42%; Unit delivery cost dropped -\$0.50
Sauce & Spoon Tablet Rollout	Retail Ops	Asana, Survey Analysis	Revenue increased >20%; Checkout time <1 min	Food waste reduced 50%; CSAT jumped from 72% to 86%
Virtual Verde Scrum	Agile / Scrum	Scrum, Burndown Charts	Delivered 6 of 8 prioritized high-value sprint items	Managed mid-sprint vendor system critical fix
Tableau Portfolio	Data Viz	Tableau Desktop, Public	Published 20 enterprise-grade dashboards	Covered \$233M+ revenue datasets and 144K+ accident records